

15740
SAR2-D2

DECODE 2025-2026

TEAM

SAR2-D2

FTC Team 15740, SAR High School

We are a part of the SAR Robotics Team, which has grown to 30 members and contains two other teams (HiveMind and Paranoid Androids on Steroids). This is our seventh year participating in the FIRST Tech Challenge, and we are excited to grow and meet new people who share our love of STEM and learning!

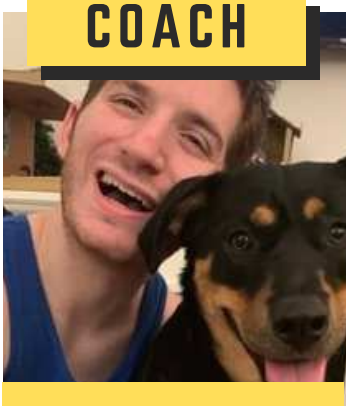
Future FTC Captain!

COACH



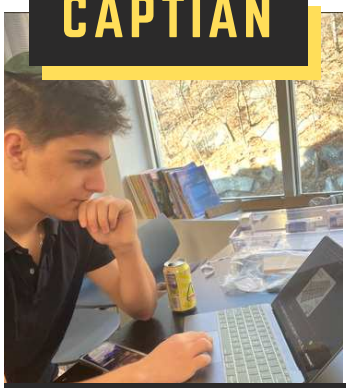
MR. CASO

COACH



MR. DARDIK

CAPTIAN



ILAY G.

CAPTAIN, CODE GENIUS



ARON K.

CAPTAIN, BRINGER OF SNACKS



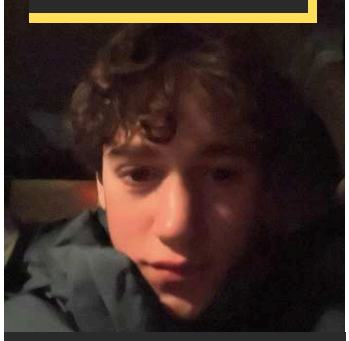
HILA L.

HARDWARE, 3D PRINTING MACHINE



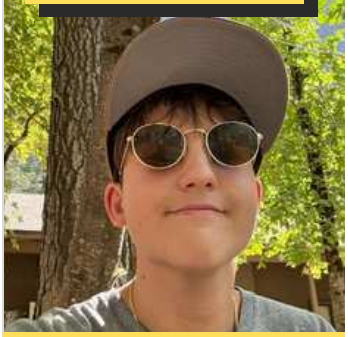
DANE E.

HARDWARE



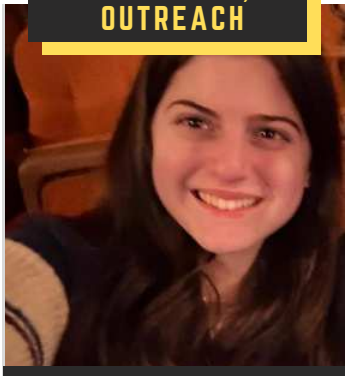
GAVI M.

HARDWARE, SOFTWARE DABBLER



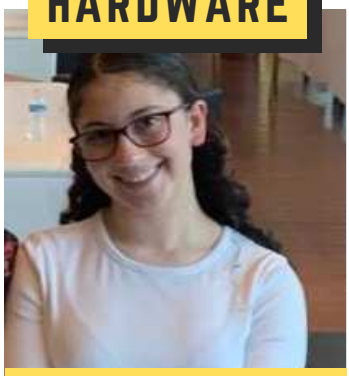
ASHER L.

HARDWARE, OUTREACH



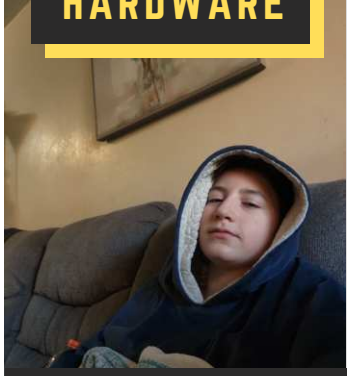
SYDNEY M.

HARDWARE



TALYA M.

HARDWARE



BINYAMIN W.

Season Goals

Our goal as the SAR2-D2 team is not only to do well in the FIRST Tech Challenge but also to build a community of engineers with a shared interest in robotics.
Whether it's through working with our sister teams, planning lessons for a robotics club, or talking to other students who are similarly passionate about STEM, we are determined to get people excited about robotics. This season will be a successful one if we manage to make people smile at the idea of engineering, spreading the joy that we all find in building our robot.

Sustainability Plan

Our team comprises of two freshmen, three sophomores, one junior, and three seniors, all playing an equal role in the design and building process. We have worked hard to foster an environment of creativity and learning within our team.

The SAR Robotics Team has grown dramatically, having 40 applicants and only 10 spots to fill this year. We've added more teams for three (SAR2-D2 and our sister teams, 19167 Robosapiens and 20789 Paranoid Androids on Steroids). Still, due to the rapidly growing interest, SAR has also created an intramural robotics league mentored by our three FTC teams to allow students to practice their engineering skills and prepare to join the Robotics Team the following year.

We encourage people to apply to the team through several means. Firstly, we volunteer at our school's open house at the beginning of the year for new parents and students, in which we represent our team and show off our robot. Secondly, through our outreach to our middle school robotics team, which we did last year and will do this year, we encourage the students to try out for our FTC team in the upcoming years when they enter high school. Both of these methods, in addition to sending out emails and hanging up posters around our school, have successfully encouraged people to try out for our team.

Business Plan

Throughout the 2025-2026 season, we have maintained careful records of all our transactions and ensured to stay under our **$1000 budget** alongside money received from our **sponsors**. The SAR Robotics Team has an overall budget of $3000, divided equally among three squads. This year, we already had numerous parts from the previous year after switching from REV parts to goBilda parts, and 3D printed several parts of our robot. This year, the amount we spent was much lower than previous years since we only needed to buy a few small parts.

Sponsors

Donations:

Kai Fan

Riverdale Judaica

riverdale kosher market

MOSS CAFÉ

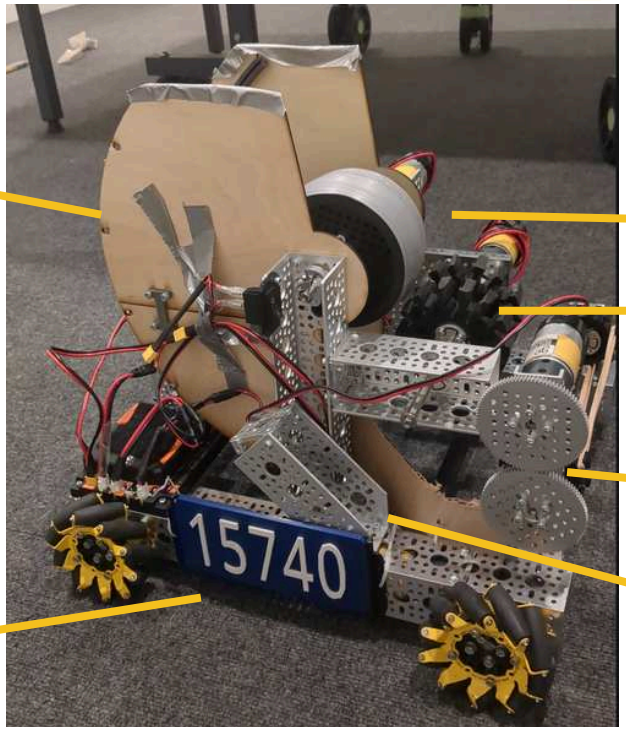
ROLEN BAGELS

Website Discounts:

REV ROBOTICS

goBILDA DREAM • MAKE • REPEAT

ROBOT OVERVIEW



"The Hood"

Propelling outtake

Transfer Gears

Custom printed gears

Battery Holder

Chassis

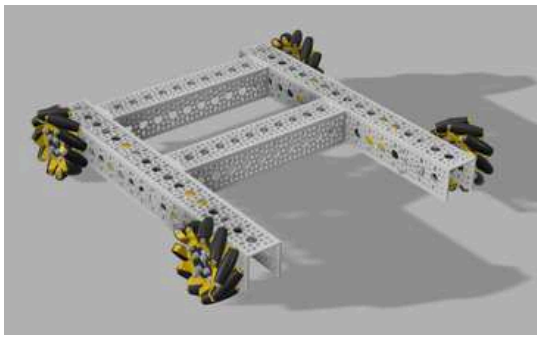
Robot Plans And Goals:

- **Minimize Iterations** - Using Fusion 360, we were able to minimize the number of iterations of each robot component by modeling them before we built them. Using Fusion enabled us to make custom laser cut and 3D printed parts that need to be precisely measured to fit on the robot(like the hood).
- **Efficiency** - We tried to make our robot as efficient as possible this year, alongside being functional.
- **Simplicity and Reliability** - We focused on making our robot as simple as possible (while keeping in mind efficiency) to not only help our drivers master control of our robot but also to avoid artifacts falling out of our robot while transferring them from the intake to outtake, and making them easier to collect.

REV vs. goBilda

In past years, we have used REV parts to build our robots. This year, however, we decided to use primarily goBilda parts for a few reasons. We found that they were much more reliable, durable, and were overall better for around the same price.

CHASSIS



CAD: Chassis

For our chassis, we used **four mecanum wheels, each powered by goBilda 312 RPM Yellow Jacket motors** (tucked into the side channels of the chassis and connected using a 1:1 gear ratio) to allow for precise omnidirectional movement since they make it easier to drive in both the Driver Controlled and Autonomous periods.
We used **goBilda's 17 and 10-hole U-channels** as the base since they are durable and prevent our robot from easily breaking when it inevitably bumps into parts of the field



goBilda Odometry Pods

For Autonomous, we wanted to utilize **RoadRunner, a trajectory-based algorithm that allows us to follow complex paths**, like splines and curves, that aren't possible with the encoders built into the motors alone. The algorithm **requires that odometry pods to be attached to the chassis**, who's dead wheels and encoders measure the distance the robot travels.
We used **goBilda's odometry pods**, which are **durable and work much better than the ones we first built in previous years**, which used REV encoders.

The part of our robot that we forget about every year is the control and expansion hubs, along with the battery always putting them on last. The problem is at that point, we barely have any space left. This year, we prioritized thinking about them earlier on in the design process, and found a support channel on our chassis that we determined would be a perfect for the battery. We slided it in and now we can confidently drive our robot around without worrying about it falling off or disconnecting from our control and expansion hubs.



INTAKE

Our intake design went through several iterations before reaching its final, reliable form.

V1

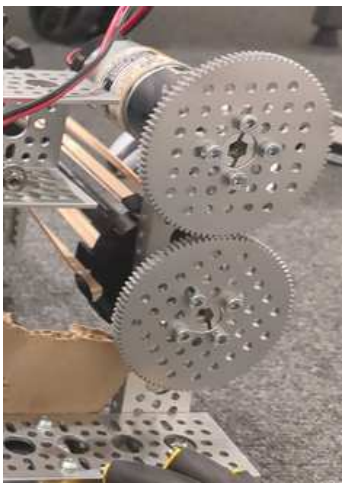
Early in the season, our first intake version focused on rigidity and simplicity. We used a **single rigid roller** mounted directly to a shaft, assuming that firm contact would be enough to pull balls in. In practice, this design struggled significantly. Artifacts frequently slipped due to inconsistent contact, and the rigid roller required extremely precise robot alignment to work effectively. Any slight offset caused the artifact to bounce away rather than be collected.

V2

In our second iteration, we attempted to improve grip by **adding textured 3D-printed rollers** on both sides of the shaft. While this increased friction, it introduced a new problem: the rollers compressed inward under load, causing binding and inconsistent rotation. Additionally, the rigid geometry still made the intake unforgiving to positioning errors, which slowed down our cycle times and increased driver difficulty.

V3

Our final intake design prioritized compliance and tolerance. We redesigned the system around a driven shaft connected to **two custom 3D-printed gears,** separated by spacers to maintain consistent alignment. Between these gears, we **mounted rubber bands that act as compliant rollers**. This compliance allows the intake to adapt to small misalignments and variations in artifact position without jamming. When spinning, the rubber bands generate controlled friction and function as a sweeping mechanism, guiding artifacts smoothly into the transfer system. This design dramatically improved reliability, reduced jams, and allowed faster, more forgiving artifact collection, aligning with our goal of simplicity and efficiency.





TRANSFER

Initially, our transfer mechanism was **tightly coupled** to the intake and outtake systems.

In **Version 1**, we **relied on timing** both systems together, spinning motors simultaneously to move artifacts through the robot. While this worked in controlled testing, it quickly proved unreliable during real driving. Any delay or variation in timing caused artifacts to either stall mid-transfer or overshoot into the outtake unpredictably. This also forced drivers to wait for full motion cycles, increasing cycle time.

For **Version 2**, we introduced a passive holding area using rigid geometry to cradle the artifact between intake and outtake. Although this reduced overshooting, it created new problems. artifacts frequently became stuck if they entered slightly off-center, and the lack of compliance meant that jams were difficult to recover from mid-match. Additionally, the system still required precise synchronization with the outtake motor, limiting flexibility in where and when we could score.

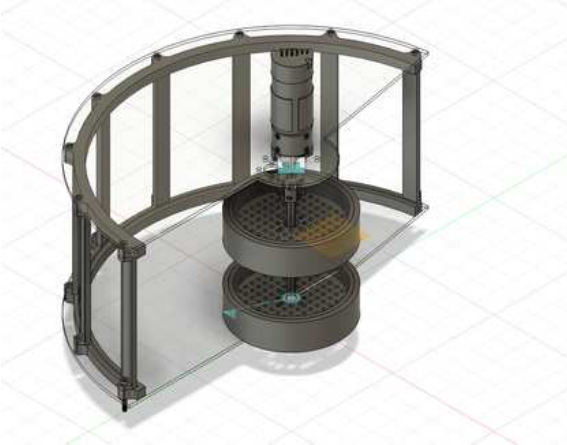
Our **final transfer** design focused on decoupling systems and **enabling asynchronous control**. We implemented a compact rubberized, gear-shaped wheel that applies **consistent friction** to hold the artifact securely in place. The outtake runs continuously at a constant speed, while the transfer wheel activates only when scoring is initiated. At the press of a single button, the transfer spins for a precisely tuned duration, feeding the artifact smoothly into the outtake. This approach eliminates repeated motor spin-ups, reduces mechanical stress, and allows scoring from flexible field positions. By separating timing logic from mechanical motion, our transfer system **became faster, more reliable, and significantly easier to control**.

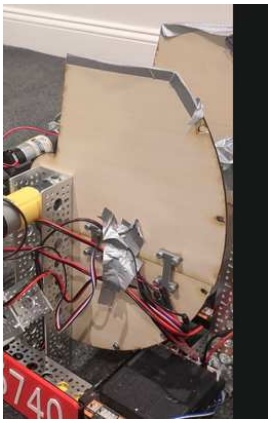
OUTTAKE

Our first outtake iteration was purely experimental and focused on basic functionality. **Version 1** used a straight channel with a servo-powered push mechanism to force the artifact upward. While simple, this design suffered from inconsistent launches. The artifact often lost contact with the channel, resulting in unpredictable trajectories and unreliable scoring, especially from longer distances.

In **Version 2**, we introduced a curved path to better guide the artifact and maintain contact throughout its motion. However, the curvature was shallow and friction was inconsistent, causing the artifact to slow down near the top or exit with insufficient velocity. We also found that repeatedly accelerating and decelerating the outtake motor introduced timing variability and unnecessary mechanical wear.

Our **final outtake** design refined both geometry and motion control. We CADed a semi-circular channel, which we laser cut and reinforced with 3D-printed components for accuracy and durability. The curvature ensures constant contact, while carefully tuned friction allows smooth acceleration without stalling. At the midpoint of the arc, a servo applies additional force, propelling the artifact toward the exit. The channel was optimized to launch artifacts at a 60-degree angle, which we determined provided the best balance between height and distance. Increased friction near the exit ensures rapid acceleration and consistent launch velocity. Keeping the outtake motor running continuously allows instant feeding from the transfer system, enabling fast, repeatable scoring from known positions on the field.





CODE

Code Structure



This year we restructured our code to make it **easier to read and learn**, which was useful when **teaching our rookies this year how to code**. We have a **Hardware** class which contains all our hardware objects as well as useful functions the robot uses during both the Autonomous and the Driver-Controlled periods. We put all our robot-specific values into a **Constants** class that we could import whenever needed. Finally, we have both **TeleOp** and **Autonomous** OpModes to run during the competition.

Autonomous Code

We have significantly emphasized our autonomous code this year and have used many different methods to control our robot precisely, independent of external factors.

Road Runner

Our robot uses a motion profiling library called Road Runner to follow complex paths and trajectories.

- Using odometers rather than drive encoders, which we used in previous years, we can track the position of the robot at all times
- Using different functions, we can have our robot follow complex paths
- We used Meep Meep to visualize our autonomous before uploading it to our robot

Asynchronous Code with FSMs

While driving, our robot needs to control the intake, transfer, and outtake asynchronously (meaning without pausing the code until finishing)

- We used Road Runner's Asynchronous Trajectory functions to drive while being able to control other hardware objects
- We designed and tuned asynchronous PID Controllers to control our outtake extremely accurately
- We used an FSM (Finite State Machine) to chain different asynchronous trajectories together to create a working autonomous

Sample Async Trajectory:

```
Trajectory sampleTrajectory = drive.trajectoryBuilder(startPose)
        .lineToLinearHeading(new Pose2d( x: 0,  y: 20, Math.toRadians(-90)))
        .build();
drive.followTrajectoryAsync(sampleTrajectory);
```

CODE CONTD.

TeleOp Code

This year, we implemented new techniques for our Driver-Controlled period to allow our robot to move smoothly and efficiently through the field.

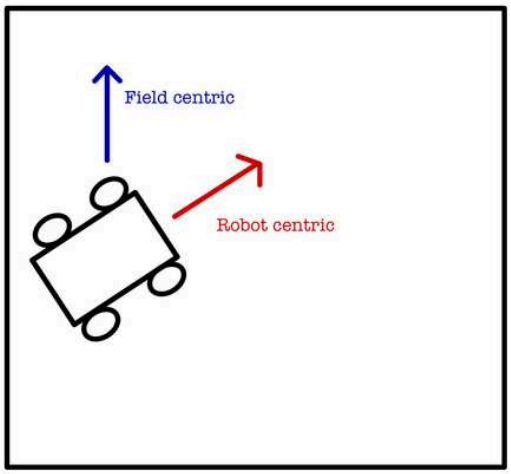
Field Centric Drive

We used a field-centric drive this year as opposed to a robot-centric drive.

- Lets us turn and drive simultaneously
- Makes it easier to pick up and score samples and specimens
- Feels more intuitive and is easy to learn

We found that we were having problems with ESD (Electrostatic discharge) from the field which was causing our internal gyroscope to mess up. To fix this issue we added a button which re-initializes our gyroscope whenever it breaks which fixes our Field-Centric drive.

The robot moves relative to the driver rather than the robot (In this case, the driver is holding forward)



Pitch Control of Intake

To allow us to control our intake smoothly, we used a technique commonly used in video games to handle movement speeds with different framerates

```
public void rotateIntake(double speed, ElapsedTime deltaTime) {
    intakePitchVal += speed * deltaTime.time();

    constrain(intakePitchVal, min: 0, max: 1);

    intakePitch.setPosition(intakePitchVal);
}
```

Single Button Automation

- To make transferring from our intake to outtake more efficient we created an automatic, asynchronous transfer system that, at the click of a button, activates the transfer while allowing us to drive concurrently

```
public void updateTransfer() {
    telemetry.addData( caption: "Status", value: "Transferri
    telemetry.update();

    if (time.milliseconds() <= 750) {
        robot.setIntakePitch(0.3);
    }
    else if (time.milliseconds() <= 1750) {
        robot.intakeWheel.setPower(-INTAKE_WHEEL_SPEED);
    }
    else if (time.milliseconds() <= 2500) {
        robot.intakeWheel.setPower(0);
        robot.setIntakePitch(0.55);
    }
    else if (time.milliseconds() <= 3000) {
        robot.setOutakeTarget(5.0);
        robot.updateOutake();
        robot.setBucketPos(0.4);
    }
    else {
        isTransferring = false;
    }
}
```

OUTREACH

SAR STEM DAY



Over forty families participated in SAR Academy's Family STEM Day, an event designed to engage the community with STEM education in a **fun and interactive way**. The day featured a Chanukah-themed program with a variety of hands-on activities, including coding challenges, a dreidel design competition, robot programming, and more.
Our team volunteered at several STEM stations, assisting with setup, and guiding **participants of all ages** through the activities. We worked alongside SAR Academy's teachers and other student volunteers to ensure the **smooth operation** of each station.



Family STEM Day provided a unique opportunity for parents, children, and even grandparents to collaborate and learn together. Dr. Chaya Fine, SAR Academy's Director of Science Curriculum & STEM Initiative, noted the event's success in sparking curiosity across many generations.
By participating in Family STEM Day, we **strengthened our connection with the community** and **supported the promotion of STEM education for all ages**. The event showcased the power of collaboration and the joy of learning, values that align with our team's mission to inspire and empower others through STEM.

SUPPORTING FTC TEAMS

One of our team's core principles is sharing our knowledge with others. This year, we utilized our expertise in **software and hardware integration** to develop functional systems to support **numerous** FTC teams with calculations and coding to control their robot components. For example, we assisted one team in designing a PID control system to maintain a linear slide, attached to a single-jointed arm, at a stable height. Our efforts have extended both in person and through the FTC Discord community!

SHAPING ACADEMY

As part of the SAR FTC team, our **main goal** is **teaching** and **inspiring** children of **all ages** about **robotics and STEM**. One way we fulfilled this goal is through partnering with **Shaping Academy**, a **special needs school in Miami**. We have **contacted** that school weekly, where they work through the **guided lessons** we provide them, and then once a month, we teach them face-to-face on Zoom.

Two years ago, we started by hosting a "**STEM Day** for them at **their school**, where we taught them to build basic **wooden race** cars. The **students loved it** so much that the **school asked** if we could **continue teaching** their **2nd** to **5th graders**. We then offered them to start a **Robotics club**, where we would create **lessons free** of charge. (provided they buy the robot kits for this activity). With the help of their occupational therapists and kind principal, we began three weeks **later with our club as part of their weekly schedule**.

Our robotics team spent a few hours weekly **making thorough lessons that teach, entertain, and challenge the kids**. We also met monthly over **Zoom** to personally interact with the children and teach them. These lessons are based on the **MakeBlock mBot2 robot**, which has many different **hardware and software capabilities**, perfect for kids their age.



We taught them how to drive the robot using **loops**, **variables**, **control statements**, **sensors**, **etc**. At the end of every lesson, we provided them with **challenges created to develop their critical thinking skills**. These **challenges encompass** all that we have taught them in the **current lesson, as well as previous ones**.

This year, although we didn't need to create any new lessons, we continued to support the **Robotics Club** for **Shaping Academy** for the new grade of students. **We aim to spread our STEM knowledge to everyone and empower future coders and engineers.** We're looking forward to continuing this in the future!







FUTURE PLANS

Some of our future outreach plans include but aren't limited to:

- Meeting with other teams to learn strategies and how to improve our robot and team this season and in future seasons. We have previously used Instagram as a method of communication with other teams, and we will continue to use it to continue meeting new teams!
- Doing an exchange with our school's middle school branch's FLL team - we will go to see what they are doing and help them (like we have done in past years), and then have them visit us and our robots, with the hope that we will inspire them to continue robotics and engineering in high school, like joining our FTC team.
- Making FTC coding videos on basic and complex topics for rookie and returning teams, such as how to use RoadRunner and PIDs, or how to 3D model a robot in Fusion 360
- Volunteering at upcoming FTC and FLL competitions
- And more!